

Data Analytics (DA)

Module -1 - Excel

Excel Assignment 1 – Data Exploration

Dataset: Product Dataset

Attributes: Product ID, Product Name, Brand Name, Quantity, Category, Price

Basic Data Exploration

1. Sum, Count, and Average

Perform the following calculations using Excel functions:

- Calculate the total price of all products in the dataset.

Formula : =SUM(D2:D35)

Result : Rs.10100

29-SEP-CA	Smartphone	Google	800	45	Electronics
03-JUN-CA	Sunglasses	Ray-Ban	130	25	Fashion
11-JUL-CA	Blender	Vitamix	400	40	Kitchen
16-APR-ES	Headphones	Bose	230	20	Electronics
07-MAR-CA	Dress	Zara	60	30	Fashion
13-APR-CA	Toaster	Hamilton	40	10	Kitchen
24-MAY-CA	Fitness Tracker	Garmin	130	5	Electronics
02-DEC-CA	Jeans	Levi's	50	50	Fashion
17-JUN-IN	Laptop	HP	950	25	Electronics
09-JUL-FR	Watch	Casio	100	20	Accessories
		Total	10100		

- Determine how many products exist in the dataset.

Formula: =COUNTA(UNIQUE(Table1[Product Name]))

Result: 21

- Calculate the average price of the products.

Formula: =AVERAGE(D2:D35)

Result: 297.06

16-APR-ES	Headphones	Bose	230	20	Electronics
07-MAR-CA	Dress	Zara	60	30	Fashion
13-APR-CA	Toaster	Hamilton	40	10	Kitchen
24-MAY-CA	Fitness Tracker	Garmin	130	5	Electronics
02-DEC-CA	Jeans	Levi's	50	50	Fashion
17-JUN-IN	Laptop	HP	950	25	Electronics
09-JUL-FR	Watch	Casio	100	20	Accessories
		Total	10100		
		Average	297.06		

2. Minimum and Maximum Values

Using Excel functions:

- Identify the minimum price among all products.

Formula: =MIN(D2:D35)

Result: 30

- Identify the maximum price among all products.

Formula: =MAX(D2:D35)

Result: 1000

24-MAY-CA	Fitness Tracker	Garmin	130	5	Electronics
02-DEC-CA	Jeans	Levi's	50	50	Fashion
17-JUN-IN	Laptop	HP	950	25	Electronics
09-JUL-FR	Watch	Casio	100	20	Accessories
		Total	10100		
		Average	297.06		
		Minimum price	30		
		Maximum price	1000		

Logical Function – IF

- Create a new column named Price Range.
- Use the IF function to categorize products based on their price:
 - If Price $\geq$ \$500 $\rightarrow$ classify as High Price
 - If Price $<$ \$500 $\rightarrow$ classify as Standard Price

Formula : =IF([@[Price (\$)]] >=500,"High Price","Standard Price")

=IF(D2 >=500,"High Price","Standard Price")

Product ID	Product Nam	Brand Nam	Price (\$)	Quantity	Category	Price Range
28-JAN-US	Laptop	Dell	1000	30	Electronics	High Price
15-FEB-US	Sneakers	Nike	80	15	Fashion	Standard Price
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	Standard Price
11-APR-US	Smartphone	Samsung	900	25	Electronics	High Price
22-MAY-US	Backpack	North Face	70	20	Outdoor	Standard Price
07-JUN-UK	Headphones	Sony	200	45	Electronics	Standard Price
19-JUL-UK	T-shirt	Adidas	30	5	Fashion	Standard Price
23-AUG-UK	Blender	Ninja	90	35	Kitchen	Standard Price
05-SEP-UK	Tablet	Apple	500	50	Electronics	High Price
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	Standard Price

4.Conditional Functions – SUMIF and COUNTIF

Perform the following conditional calculations:

- Use the SUMIF function to calculate the total price of products in the Electronics category.

Formula: =SUMIF(Table1[Category],"Electronics",Table1[Price (\$)])

Result: 8050

- Use the COUNTIF function to determine the number of products with a price less than \$100.

Formula: =COUNTIF(Table1[Price (\$)],"<=100")

Result: 12

17-JUN-IN	Laptop	HP	950	25	Electronics	High Price
09-JUL-FR	Watch	Casio	100	20	Accessories	Standard Price
		Total	10100			
		Average	297.06			
		Minimum price	30			
		Maximum price	1000			
		SUMIF	8050			
		COUNTIF	12			

5. Text Formatting – LEFT, RIGHT, MID

- Use text functions to extract information from the Product ID column.

- Create the following new columns:

- Create a new column named Day with the first 2 characters of each 'Product ID' using the LEFT function.

Formula: =LEFT([@[Product ID]],2)

- Create a new column named Country Code by extracting the last 2 characters from the 'Product ID' column using the RIGHT function.

Formula: =RIGHT([@[Product ID]],2)

- Create a new column named Month by extracting the 4th to 6th characters from the 'Product ID' column using the MID function.

Formula: =MID([@[Product ID]],4,3)

Country Cod	Day	Month
US	28	JAN
US	15	FEB
US	03	MAR
US	11	APR
US	22	MAY
UK	07	JUN
UK	19	JUL
UK	23	AUG
UK	05	SEP
UK	14	OCT
IN	17	JUN
AU	25	NOV
DE	08	DEC
CA	18	FEB
ES	16	APR
CA	21	AUG
CN	20	AUG
IT	27	JAN
UK	01	MAR
US	14	AUG